University of Southampton

Data Analytics and Organisational Decision Making

Contents

1. Introduction

This report will be divided into the following two parts: the first part will discuss the challenges that Bank X has faced or is likely to face in implementing data analytics and the impact of big data analytics on its decision making based on the given text. The second part will be the analysis and summary of the provided dataset through Power BI.

2. Part I QA: Data Analytics Implementation at Bank X

2.1 Key challenge, impact and their critical analysis

1. Problems encountered in data quality and integration

There are difficulties in data legacy, transfer and cleaning. Data fragmentation, inconsistent formats, absence or repetition led to deviations in analysis results. The challenges of data legacy, transfer and cleaning: Ismail, the managing director of BankX, has a particularly profound observation: "These systems can perform individual tasks excellently, but when it comes to integrating data, they become a nightmare." This fragmentation leads to two direct consequences: one is the information loss during the data migration process, and the other is the huge cost required for cleaning and converting data. What is more troublesome is that maintaining these systems requires regular downtime, further delaying the transformation process.

After 40 years of operation and multiple mergers and acquisitions, BankX has accumulated dozens of IT systems from different eras in its banking system, each carrying the business logic and data standards of a specific historical period. These systems are superimposed layer upon layer like archaeological strata. What's more complex is that each merger and acquisition bring a complete IT ecosystem to the acquired party, and there is a large amount of repetitive but not completely consistent data among these systems.

These difficulties have jointly led BankX's current decision-making practice into a "dilemma": if it insists on comprehensive data cleaning, the project may not show results for three years. If the compromise solution of "good enough is fine" is adopted, it will affect the accuracy of the subsequent analysis. Data quality issues are not only a challenge for technical cleaning but also expose the lack of a unified data governance framework in enterprises. Data analysis teams must spend months on data cleaning and transformation, which not only increases costs but also makes it impossible for banks to respond quickly to market changes. Long-term data quality issues also have an impact on decision-making. Due to the unreliability of historical data, the management gradually developed a lack of trust in the results of data analysis and instead relied more on personal experience and intuitive judgment. This trend has led the decision-making model of banks to retreat from data-driven to experience-driven, making them particularly conservative in terms of strategy formulation and business innovation. More seriously, it is the decision-making mistakes caused by data errors. When key business data of banks, such as account balances and transaction records, have errors during the system migration or integration process, it may trigger a series of chain reactions. Incorrect customer credit scores may lead to mistakes in loan approval, thereby increasing the risk of bad debts.

2. Problems encountered in the field of technology and skills

The shortage of skills is a major factor affecting enterprise data analysis. However, it is worth noting that outdated organizational forms or slow update speeds are also significant challenges in data-driven transformation. *Just as Siti Aminah, the data governance manager of*

BankX, described, "The huge legacy system is not only a technical issue, but also reflects the solidified state of organizational cognition and the allocation of rights and responsibilities.”

(Siew, E.-G., & Farouk, F. M. 2023)

The dual predicament of technology and organizational form is profoundly influencing the decision-making practices of enterprises: The complexity of technology selection leads to frequent mistakes in the matching of tools and business requirements for enterprises, and the insufficiency of team capabilities further amplifies this risk. Blindly chasing popular technologies leads to misallocation of resources. For instance, a certain auto parts factory invested 2 million US dollars in an AI predictive maintenance system, but it became a mere decoration due to insufficient data sampling frequency and the lack of skills in the maintenance team. This not only caused direct economic losses but also made subsequent technological investment decisions more conservative.

The gap between the speed of technological iteration and the organizational adaptability creates a decision-making dilemma: To maintain competitiveness, new technologies need to be adopted, but the organizational learning curve is too long, and it is difficult to digest the innovative achievements. More fundamentally, the lag in organizational form has solidified a rights and responsibilities structure that is contrary to data-driven decision-making. As shown in the BankX case, the legacy system reflects departmental barriers and cognitive rigidity, which makes it difficult for data to flow across departments and leads the decision-making process into endless coordination rather than efficient judgment based on facts. This dual constraint of technology and organization not only reduces decision-making efficiency but also forms a vicious circle: the failure of technology application reinforces the conservative tendency of the organization, while the conservative organizational form further hinders the construction of technological capabilities. Eventually, the enterprise's decision-making model is locked onto the path dominated by traditional experience, gradually losing its competitive edge in digital transformation.

3. Barriers to communication and collaboration between departments

The barriers may lead to departmental data silos, employees' resistance to change, and insufficient support from management. Data silos often stem from conflicts of interest within departments (such as performance appraisal mechanisms), and the incentive structure needs to be adjusted through top-level design rather than relying solely on technological integration. Take BankX as an example. When promoting the big data analysis strategy, the collaboration barriers among departments significantly affected the effectiveness of their decision-making practices.

Firstly, the problem of data silos leads to incomplete decision-making information - due to the lack of standardized interfaces in multiple independent systems left over from historical mergers (such as the core banking system, credit card system, etc.), coupled with data protectionism caused by the departmental performance assessment mechanism (such as the retail banking department's refusal to share customer asset data), banks are unable to comprehensively analyze customer profiles. This has led to the difficulty in implementing datadriven decisions such as "precise marketing of travel insurance products" envisioned in the case.

Secondly, the inefficient cross-departmental collaboration has prolonged the decision-making cycle - the case mentioned that "it takes one year for the model to be approved for production", reflecting the game among risk control, technology, and business departments over data ownership and process norms, ultimately causing BankX to lag in decision-making when responding to fintech competition.

The deeper impact is that the insufficient support from the management has led to an imbalance in resource allocation. Despite the establishment of a data analysis team, the old department still holds budgetary dominance, resulting in slow progress in basic tasks such as data cleaning and system migration. *A team leader may help their team avoid avoidable confrontations by explicitly stating any "non-negotiable" requirements that now take precedence over other needs, such as a vital delivery with an immovable deadline. It may be useful for teams in highly dynamic work situations to hold occasional "priority check-in" conversations (Tannenbaum, S et al., 2023).* Therefore, it is necessary to unite as one and overcome the obstacles in organizational culture and collaboration, which can fundamentally solve these problems.

4. Privacy and security compliance issues

Privacy and regulatory constraints significantly hinder the implementation of big data strategies at BankX. Regulations such as the GDPR impose strict compliance obligations, where violations can lead to legal penalties, loss of user trust, and long-term brand damage. Static security measures are inadequate for evolving threats, necessitating dynamic and continuous monitoring. For example, when BankX's risk control team attempted to integrate unstructured data (e.g., social media), GDPR's "purpose limitation" clause forced the exclusion of 20% of high-value data, reducing credit model accuracy by 12%. Additionally, new legislation such as Indonesia's 2023 Financial Data Sovereignty Act forced BankX to restructure its cross-border cloud infrastructure, delaying its anti-fraud system rollout by nine months. These disruptions have prompted a shift to privacy-preserving technologies like federated learning, though migration delays risk losing competitive advantage. The risks extend beyond finance; a European hospital was fined €8.6 million for failing to sign a Data Processing Agreement, leading to unauthorized commercial use of patient data and a 20% drop in stock value. *As cyberspace increasingly impacts all sectors, cybersecurity has become central to* *privacy, economic resilience, and national security (Xu et al., 2021).* Privacy computing, smart contracts, and automated audits are essential for secure data collaboration.

2.2 Industry differentiation consideration

Different industries have different priorities, such as finance/healthcare, where compliance and security are the highest priorities and require up-front investment. *For instance, as mentioned in this article, the data analysis platform in the medical industry must support key functions such as availability, continuity, ease of use, scalability, privacy and security. Real-time data analysis is a key demand in the medical industry, and the delay problem between data collection and processing must be addressed (Wullianallur, R et al., 2014).* This is followed by retail/manufacturing, where data integration and real-time analytics capabilities may be more critical. For startups, focus on low-cost tools and fast verification, avoiding excessive infrastructure.

2.3 A strategy for prioritizing challenges encountered by the business

Enterprises need to combine business objectives, industry characteristics and resource constraints, according to the following hierarchy:

The first priority is fundamental capacity building data quality and governance, *as indicated by Wallis et al (2007) data collections are only as valuable as the data they include, and users must be able to trust the data based on the data system's integrity and inherent quality,* which requires establishing standardized data collection processes and governance frameworks to ensure the reliability of analytical inputs, deploying encryption, rights management and compliance audits, and preventing legal and reputational risks. This aspect is very important, if the data is not trusted or legitimate, the subsequent analysis will lose value or even cause a crisis.

The second priority is to invest in technology and talent, and the key is to adapt technology and choose tools that match the business scenario (such as lightweight BI tools instead of complex data lakes). Skills upgrading is also important, filling skills gaps through training or recruitment, with priority given to developing people who are both business and technical. Because technology needs to serve business needs, not vice versa; Talent is the core of continuous innovation.

The third priority is organizational and cultural transformation to break down data silos and facilitate data sharing, Transformation is not only about technological upgrading, but also a profound change in the organizational decision-making system and operation mode. The key lies in cultivating a data-driven corporate culture, breaking down departmental barriers through measures such as incorporating data quality into KPI assessment through cross-departmental collaboration mechanisms, establishing cross-departmental data teams, and implementing talent rotation plans to promote data sharing. Through the empowerment of the management, prove the ROI of data analysis with pilot projects, strive for the support of the top management and incorporate it into the strategy. Cultural change is a long-term project, but early demonstration projects can accelerate the formation of consensus. However, transformation should not be rushed. One should avoid pursuing short-term effects at the expense of long-term value.

The fourth priority, scale and continuous optimization iterative expansion, small pilot (such as marketing optimization) gradually expanded to the whole business chain. Dynamic cost management Balance cost and performance with flexible solutions such as cloud services to continuously evaluate ROI. At the same time, avoid large-scale investment and reduce risk through agile iteration.

3. Part I QB: How big data can change bank decisionmaking

According to the material, there are three areas where big data can change bank decisionmaking:

- Big data analytics for employee tagging and segmentation for targeted marketing, e.g. by analysing data on age, employment, education, income levels, savings, loans and investments, BankX can tailor its investment products to the most relevant customers such as unit trusts, e-gold investments or fixed income investments, rather than randomly marketing to all customer segments
- Big data to manage users and analyse to assist in making smarter decisions By automating the analysis of customer data, BankX can better understand its customers and make smarter decisions. For example, if BankX identifies frequent travelers, it can offer these customers a Platinum Travel Card and Travel Plus insurance benefits.
- Providing a better credit risk assessment process, for example, by evaluating data such as credit scores, repayment histories, education levels, employment information and utility payment histories, BankX can more accurately determine the creditworthiness of a new loan applicant or assess the credit risk of an existing customer.

3.1 Specific areas of Value Provision

Providing a basis for decision-making and improve decision-making efficiency ：

Banks using big data analytics can process huge amounts of internal transactional data as well as external data sources, such as transaction records, third-party credit ratings, social media feedback, etc. To more comprehensively and accurately assess a customer's credit worthiness, and to reduce the bias of subjective judgement. By quickly processing and analyzing massive amounts of data, banks can make real-time or near real-time decisions, thus significantly improving decision-making efficiency and accuracy. For example, by introducing satellite big data analytics, India's agricultural insurance avoided data distortion and political interference in traditional manual loss estimation, significantly reduced subjective judgement errors, and improved the efficiency of claims settlement (Narayan et al.,2023 [4]).

Tagging user characteristics, based on labels categorizing users, generating positive value for the business:

BankX can use customer data to enhance its marketing campaigns. In addition, in the user classification and stock recommendation system, an algorithm is applied to classify customers into gold, silver, platinum, and bronze levels based on their financial data such as annual income, insurance expenses, and rent, and at the same time, it combines social media sentiment analysis and real-time stock data to accurately recommend matching corporate stocks to users of different levels. This kind of big data label classification not only improves the relevance of recommendations and user satisfaction, but also helps enterprises optimize their customer relationship management and marketing strategies, bringing significant positive economic benefits (Praised et al.,2022 [5]) .

Data-driven Transformation of Organizational Structure and Decision-Making Processes:

In implementing big data analytics, banks have redesigned their internal organizational structure by setting up dedicated analytics departments, introducing dedicated data analytics roles and teams, and emphasizing a data-driven decision-making culture. This change in organizational structure and processes has helped the bank to use data more effectively for business decision-making. Original: Since 2018, BankX has made a number of structural changes such as introducing new activity components, role teams and processes to strengthen its data-driven transformation efforts, in addition to starting to streamline data-related activities by upgrading staff skills, integrating data into a central data warehouse and launching a sandbox incubator program. In a survey of 161 companies globally, the study found that many had dedicated big data analytics departments to strengthen big data capabilities to support organizational agility and improve market responsiveness through data alignment at the strategic and operational levels (Yeming et al., 2022 [6]) and provided operational business training to help maximize the benefits of big datamining for the company.

3.2 Possible challenges of big data transformation

(1) Difficulties in management due to long project cycle and high complexity:

System migration and regulatory compliance significantly extend project timelines, reducing banks' flexibility in responding to market demands. Big data transformations often involve cross-departmental collaboration, which introduces coordination challenges and conflicting interests, especially in the early stages. These factors result in longer project cycles and increased management pressure. For example, the Australian Stock Exchange's CHESS replacement project using Distributed Ledger Technology, launched in 2017, was delayed due to underestimated complexity and disorganized management. By 2023, only 60% of the project was completed, after spending A\$170 million, and it was eventually cancelled. This case highlights the critical impact of poor project planning and prolonged cycles on large-scale technology initiatives. To address these issues, organizations should adopt phased implementations, clarify responsibilities early, and use agile project management to enhance efficiency and reduce transformation risks.

(2) At the organizational level, big data replaces some of the positions brought about by employee resistance:

Data-driven decision-making may harm the interests of a certain portion of the positions when improving efficiency, especially the front-line execution of the positions, for example, with the improvement of efficiency leading to attrition or the automation of the organizational nodes some of the positions (e.g., customer service auditing and other positions) may be attrition, or even the position is directly cancelled. The impact of this is likely to lead to resistance from employees within the organization, who will not cooperate with the system's iteration or the use of the new system. To solve these problems, first we need to design the corresponding performance correlation, otherwise it is difficult to apply the new system. On the other hand, for the cancelled positions, we need to pay attention to the placement of employees after job replacement, including job redesign, staff transfer training, and the development of appropriate incentives (Nonye, 2022)

(3) Problems caused by personnel changes:

Personnel changes will bring many challenges to the big data transformation. In the long-term process of big data transformation, enterprises inevitably face the natural flow of personnel, new employees will also lead to the frequent replacement of the front-line big data application users, which will lead to system operation is not familiar with the problem of reduced work efficiency, to address this problem, enterprises need to retain specific product instructions in advance, set up a more standard training process to reduce the difficulty of new employees to start, to ensure the stability of the system application. Stability of the system application. On the other hand, the technical research and development level due to the transition period will be technical demand will have peaks and valleys of the characteristics of the problem easily lead to a shortage of resources. Therefore, enterprises need to plan ahead, do a good job of talent reserves and talent training, but also through the establishment of a linear system model, combined with convex optimization technology, dynamically adjust the allocation of resources, weakening the dependence of different components, thus effectively shortening the project cycle and improving the resource utilization rate (Masaki and et al., 2019) to support the everchanging resource requirements in the process of big data transformation, and to ensure the smooth progress of the transformation work. guarantee the smooth progress of the

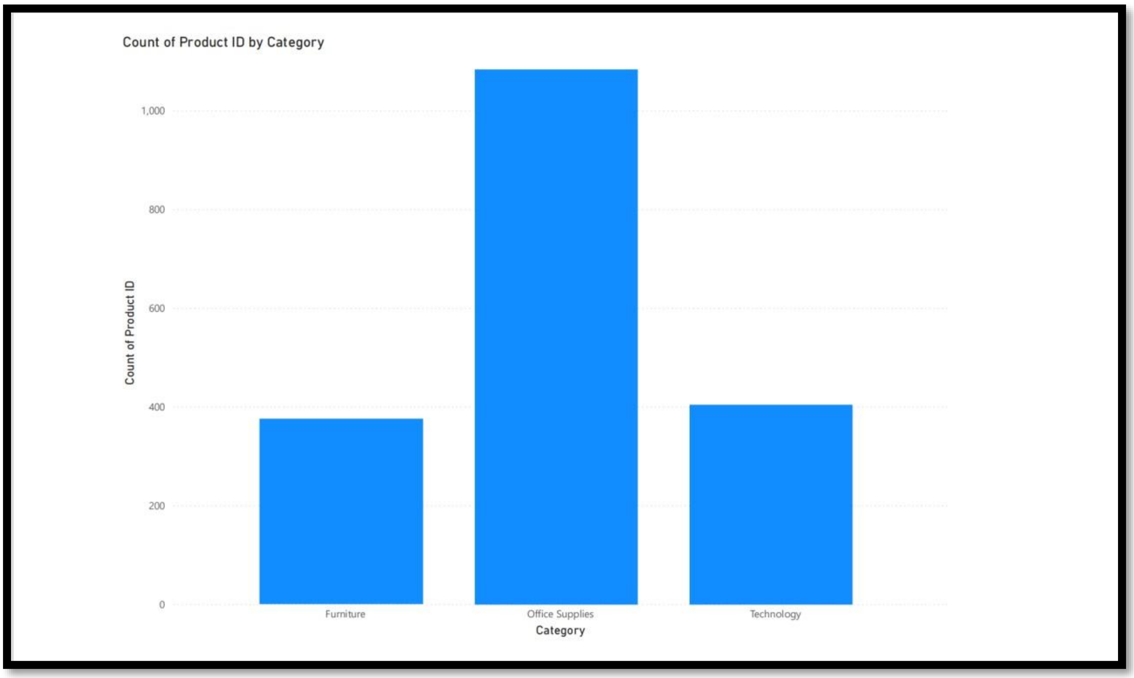
transformation.

4. Part II Question I

Introduction

This report provides a visual summary of Alpha's product and sales performance through Microsoft Power BI. The analysis covers product distribution by category, sales volume rankings, regional performance, and yearly trends in sales and profit. These visualisations highlight key patterns in demand, regional variation, and long-term business growth. The findings offer practical insights to inform product planning, regional targeting, and overall strategic decision-making.

4.1 Number of Products per Category



Visualization Insight:

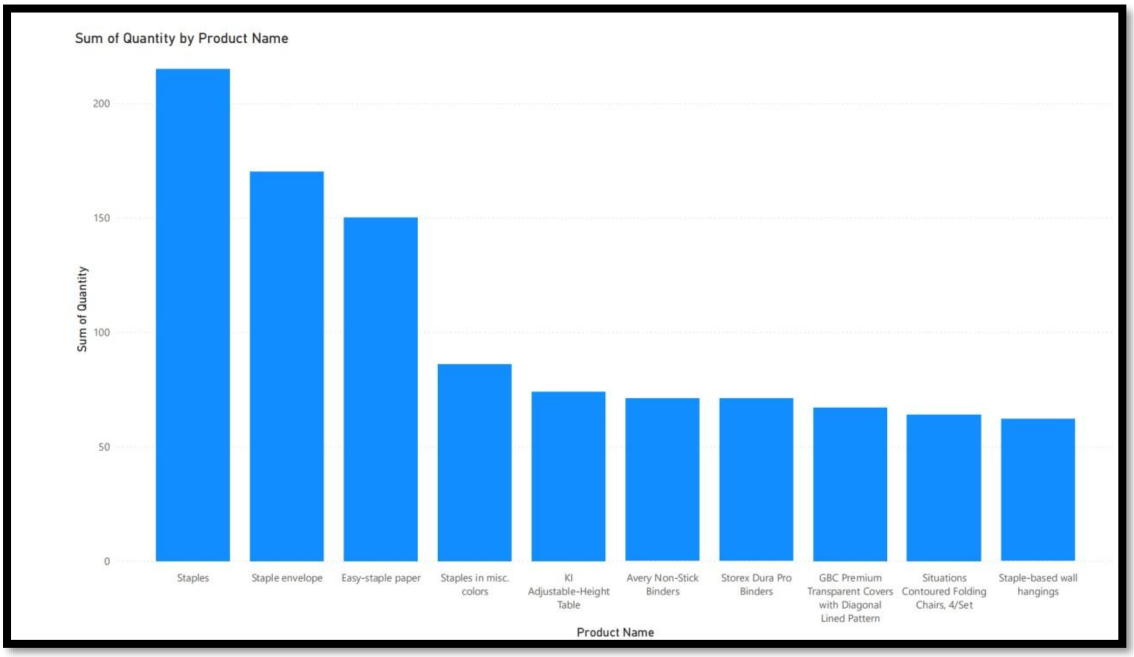
The bar chart indicates that Alpha offers three main product categories: **Office Supplies, Furniture, and Technology**. Office Supplies has the highest product count (over 1,000 items), followed by Furniture and Technology.

Business Implication:

Office Supplies are clearly Alpha's dominant product line, suggesting a strategic focus in this area.

Furniture and Technology categories could be expanded to tap into specific institutional or professional needs.

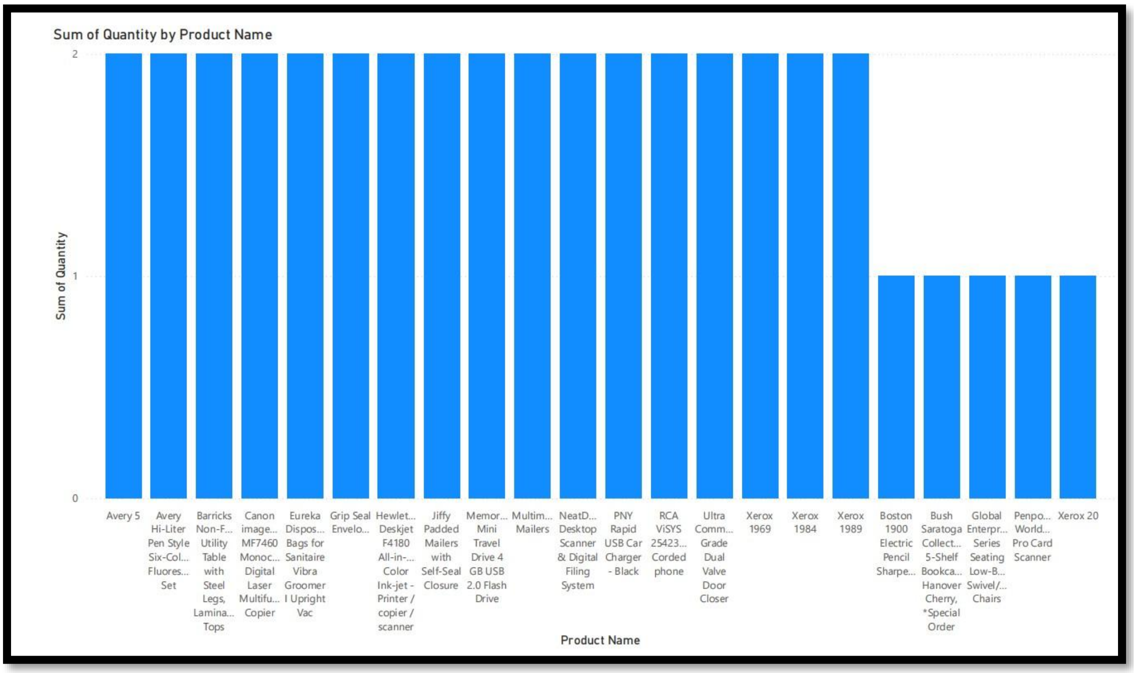
4.2 Top 10 and Bottom 10 Products by Sales Volume



Top 10 Products:

The top-selling items include **Staplers, Envelopes, Binders, Chairs, and Paper**—all essential office supplies.

Sales are heavily concentrated around frequently used, low-cost items.



Bottom 10 Products:

The lowest sellers include **USB chargers, scanners, and certain printers or laptops**, mostly tech-focused products.

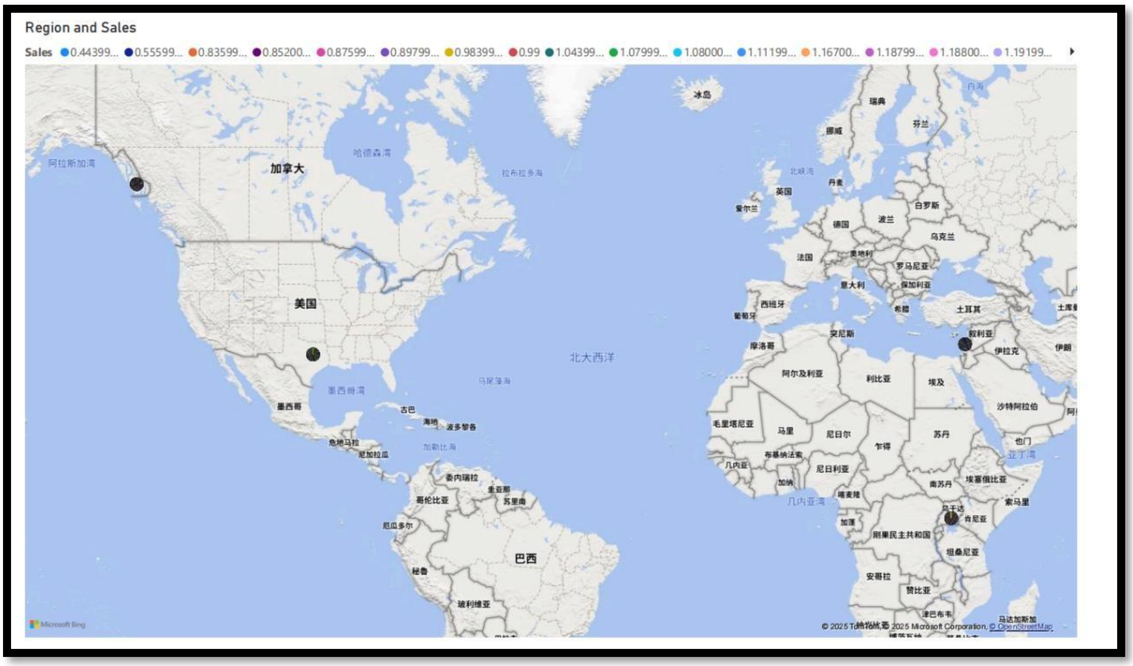
These may be low in sales volume but not necessarily low in profit margins.

Business Implication:

Alpha should enhance promotions around high-volume essentials.

Long-tail or niche products need evaluation, either reposition them or reduce lowefficiency inventory.

4.3 Regional Sales Performance



Visualization Insight:

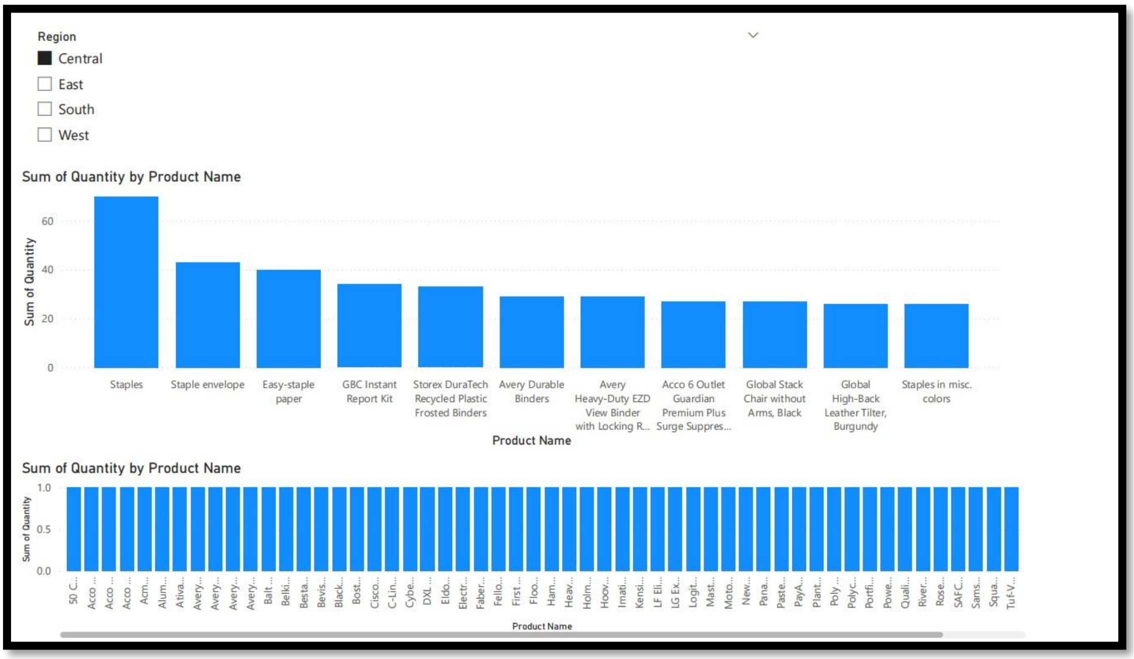
Among four regions (West, East, Central, South), **the West region leads in total sales**, followed by East and Central. The **South region lags behind**.

Business Implication:

West should be prioritized for further investment and marketing.

South may require analysis of distribution gaps or a tailored product mix to improve performance.

4.4 Top and Bottom Products in Each Region



Visualization Insight:

While the top 10 products vary slightly across regions, **Staples and Office Supplies remain dominant**.

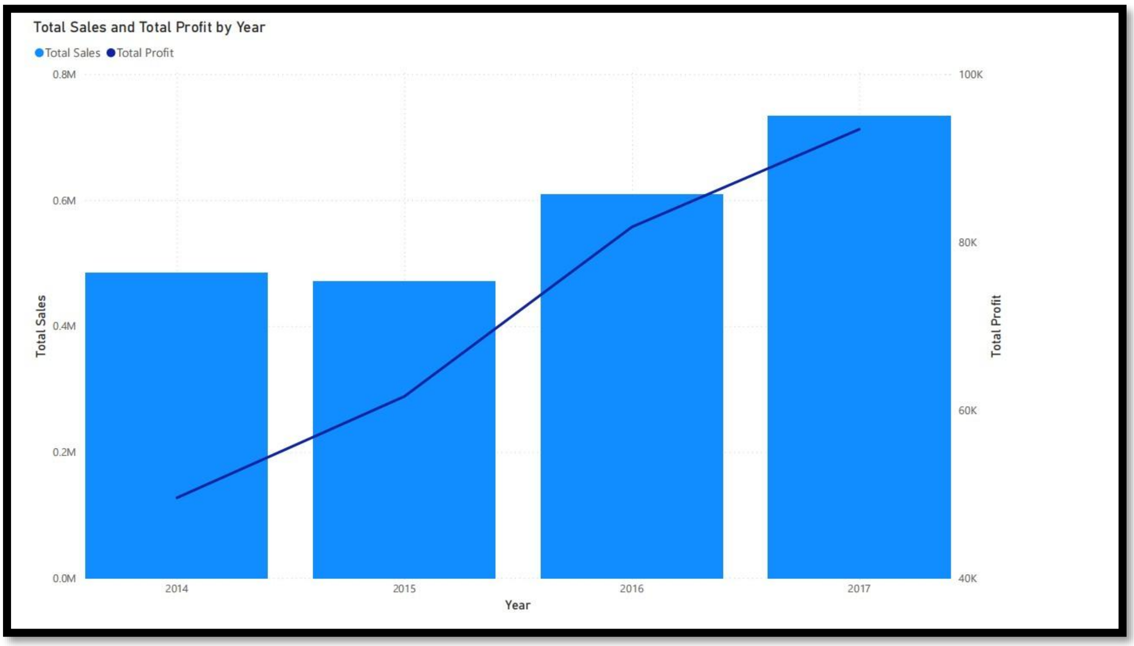
The South region's bottom products show a more limited and low-volume range, often including technology accessories.

Business Implication:

Tailored strategies by region are needed. The South could benefit from focusing on core, high-frequency items.

Consider bundling strategies or institutional targeting (e.g., schools, local offices) in lowperforming areas.

4.5 Total Sales and Profit by Year



Visualization Insight:

From 2014 to 2017, **both total sales and profits steadily increased**, peaking in 2017 (~$800K).

The profit trend closely mirrors revenue, suggesting stable cost management.

Business Implication:

Alpha is on a strong growth trajectory.

Further investments in marketing, product development, and customer segmentation could enhance this trend.

A deeper breakdown (e.g., by quarter or customer type) could support finer-grained decision making.

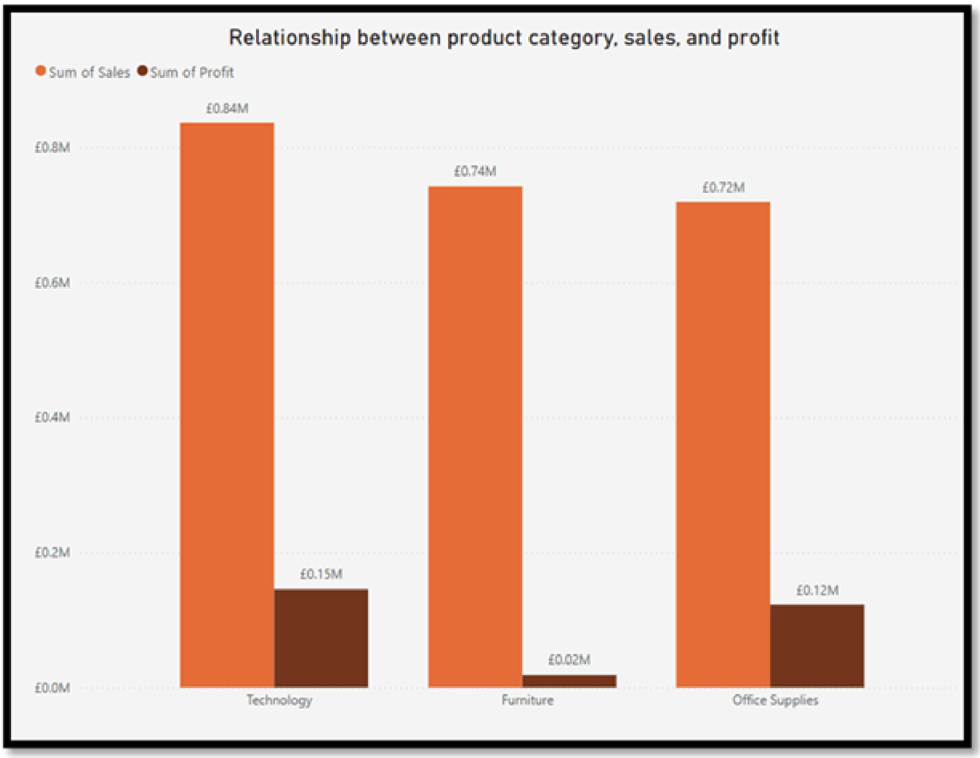
5. Part II Question II

Introduction

This represents an analytical report of Alpha Stores' performance through Microsoft Power BI. It seeks to leverage the visualisation tools to uncover performance in a number of important dimensions such as Product Categories, Customer Segments, Regional Performance, Discounting and Shipping methods. The objective is to gain an insight where business impact is derived, to aid in faster strategic decision-making and operational improvements.

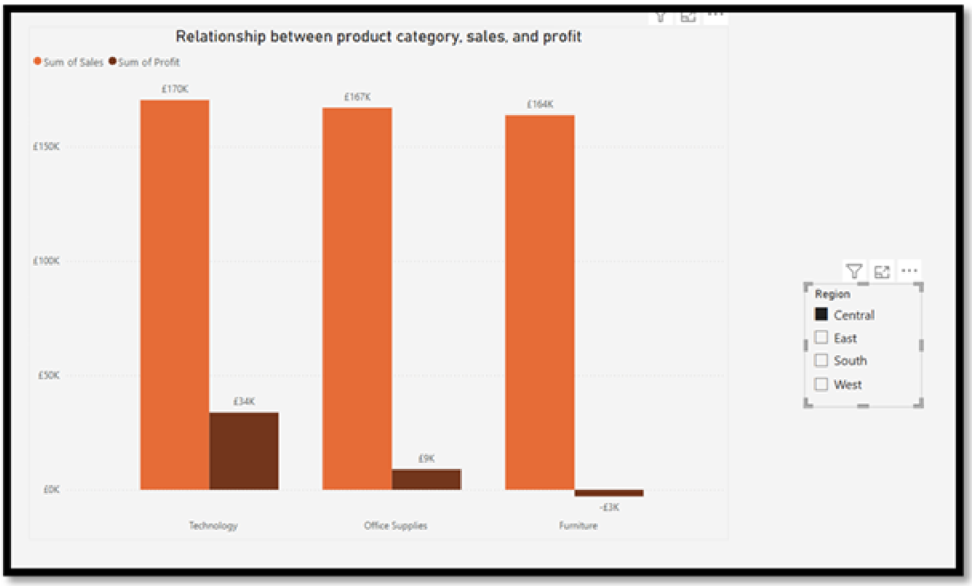
5.1 Relationship Between Product Category, Sales, and Profit

A clustered column chart with total sales and profit across the three main product categories (Technology, Office Supplies and Furniture) shows key performance differences. Technology had the largest sales (£0.84M) along with the most profit (£0.15M) generated, indicating strong revenue generation and healthy-margin performance. Office Supplies also performed strongly with sales (£0.72M) achieving solid profit (£0.12M), indicating an operationally efficient aspect of the business. When compared, Furniture had the same level of sales (£0.74M), yet generated a significantly lower-profits (£0.02M). This characteristic points toward a greater problem with margin performance, probably a combination of higher costs or discounting. The data suggests that Furniture should be getting more attention to improve sales and profitability.



5.2 Total Sales Filtered for Each Region

The bar chart displays Sales and Profits results that are filtered with a slicer by region. The analysis provides results which display a number of significant variances of sales and profit performance across the four regional markets - Central, East, South and West. Technology was the consistent leading Sales and Profit category for each of the four regions. It showed prominence in the East by £265K in sales and £47K in profit, while the technology led sales for the Central region were £170K sales and £34K profit. Office Supplies represented in the West lead in sales (£185K) and profit (£53K), even though they were sold less than the Technology. The Furniture category showed mixed performance with profitability in West , moderate profitability in East and South while being loss-leased by £3K in the Central region.


Relationship between product category, sales, and profit
Sum of Sales
Sum of Profit
£170K
£167K
£164K
£150K
£100K
£50K
£34K
£9K
£0K
-£3K
Technology
Office Supplies
Furniture
Region
Central
East
South
West


Relationship between product category, sales, and profit
Sum of Sales
Sum of Profit
£265K
£250K
£208K
£206K
£200K
£150K
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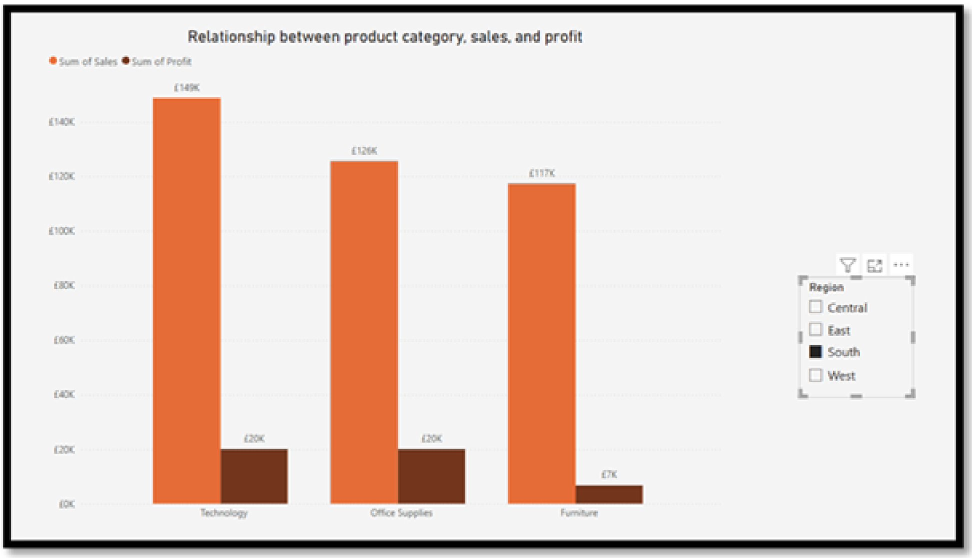
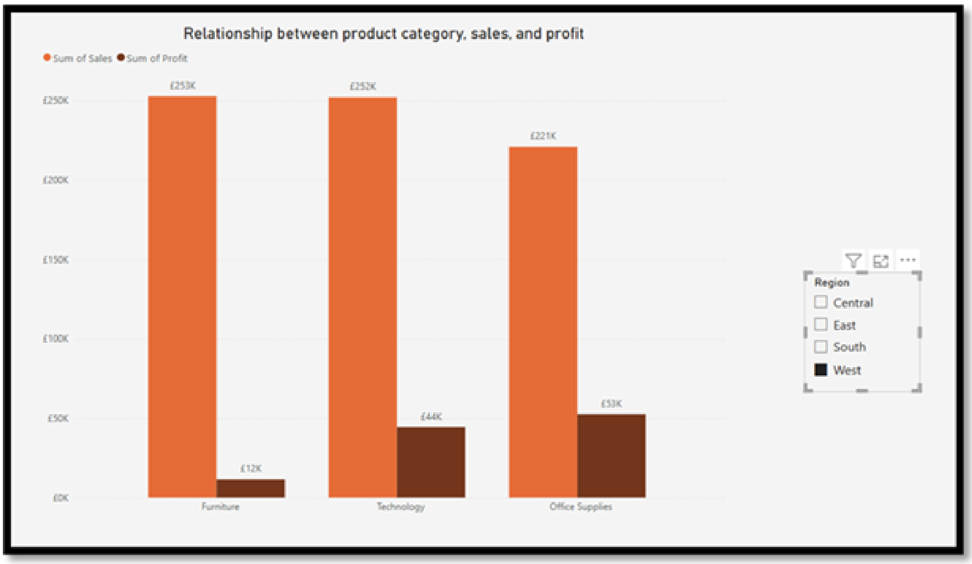
Furniture
Office Supplies
Region
Central
East
South
West





5.3 Sales, Profit, and Profit Rate by Category and Sub-Category

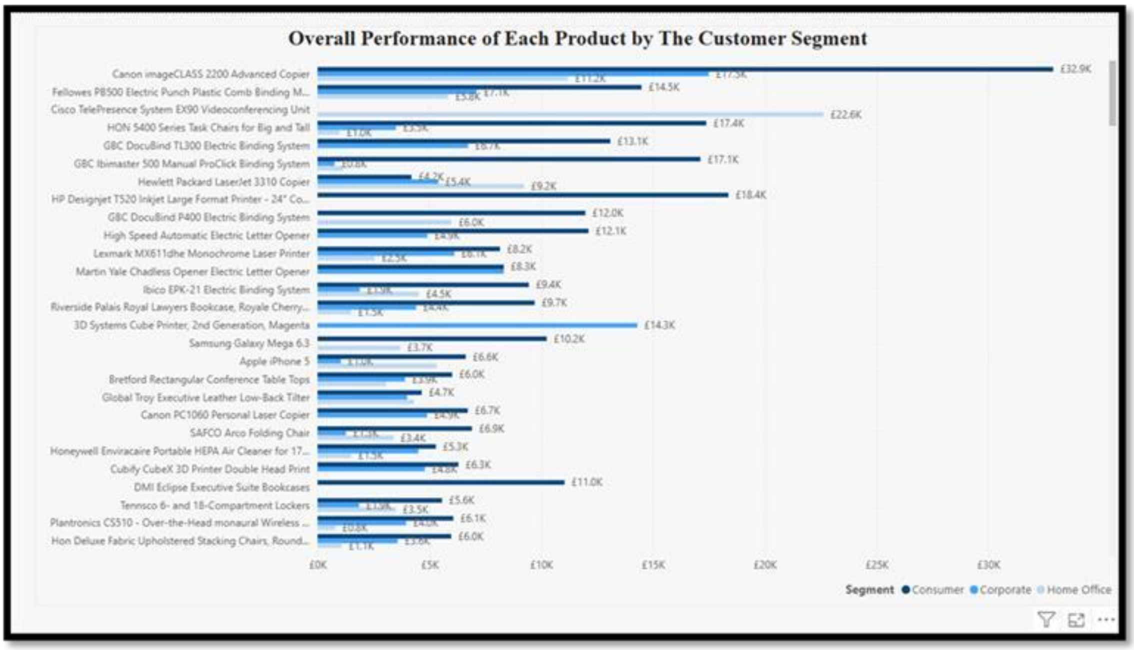
To summarize sales, profit, and Profit rate (Profit / Sales) by category and sub-category, the results are set up in a matrix visual presented by a clustered bar chart with conditional formatting. The analysis results indicate that Technology categories have the highest sales (£836K), and profit (£145K) while sub-categories like Copiers (0.37) and Accessories (0.25) have particularly high profit margins. Office Supplies performs well in high sales (£719K), and had an overall profit rate (0.17) but at the sub-category level, the results varied. The subcategories-Labels, Envelopes, and Paper had profit margins, but Supplies and Art incurred losses. Furniture categories despite high sales (£742K), had lowest overall profit (£18K) with a very low profit rate of 0.02. Sub-categories have significant losses in Tables and Bookcases. These results suggest that Alpha should trim product portfolios through the important subcategories of Technology and Office Supplies with the highest margins. It should reassess their pricing, costs, or demand issues within the Furniture category.

Category	Sum of Sales	Sum of Profit	Profit Rate
Office Supplies	**£719,047.03**	**£122,490.80**	**0.17**
Fasteners	£3,024.28	£949.52	0.31
Labels	£12,486.31	£5,546.25	0.44
Envelopes	£16,476.40	£6,964.18	0.42
Art	£27,118.79	£6,527.79	0.24
Supplies	£46,673.54	-£1,189.10	-0.03
Paper	£78,479.21	£34,053.57	0.43
Appliances	£107,532.16	£18,138.01	0.17
Binders	£203,412.73	£30,221.76	0.15
Storage	£223,843.61	£21,278.83	0.10
Furniture	**£741,999.80**	**£18,451.27**	**0.02**
Furnishings	£91,705.16	£13,059.14	0.14
Bookcases	£114,880.00	-£3,472.56	-0.03
Tables	£206,965.53	-£17,725.48	-0.09
Chairs	£328,449.10	£26,590.17	0.08
Technology	**£836,154.03**	**£145,454.95**	**0.17**
Copiers	£149,528.03	£55,617.82	0.37
Accessories	£167,380.32	£41,936.64	0.25
Machines	£189,238.63	£3,384.76	0.02
Phones	£330,007.05	£44,515.73	0.13
Total	**£2,297,200.86**	**£286,397.02**	**0.12**

5.4 Product Performance per Customer Segment

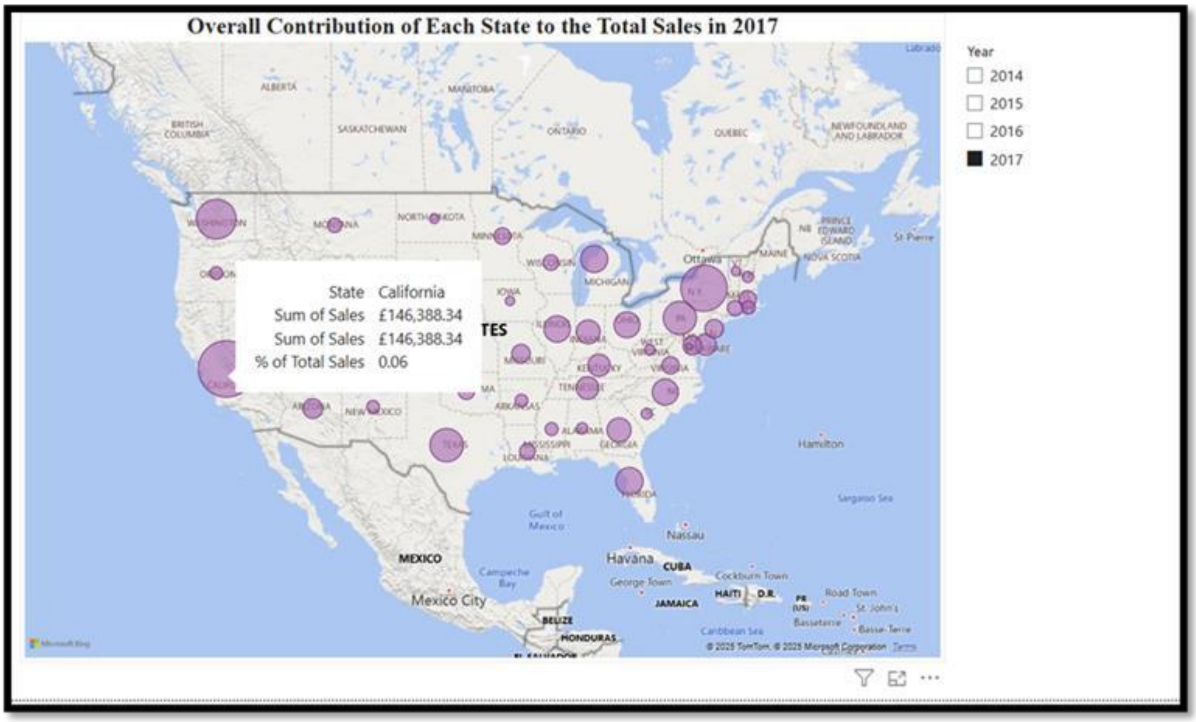
A stacked column chart compares the overall product performance for customer segments—Consumer, Corporate, and Home Office. It displays products, like the Canon imageCLASS 2200 and Fellowes binding machines which have significant sales across all three customer segments, with Consumer and Corporate segments contributing the largest share of sales.

Majority of listed products show £0 sales indicating that they did not sell or were less popular with large segments of customers with the take away being Consumers and Corporate segments account for almost all of the segment sales. The Home Office segment has occasional spikes for the items, but infrequent sales. Overall, this information can guide Alpha on how to market towards the segments where they have a stronger uptake on products, helping for more initiative consideration as well as inventory planning for different segments.



5.5 Contribution of Each State to Total Sales in 2017

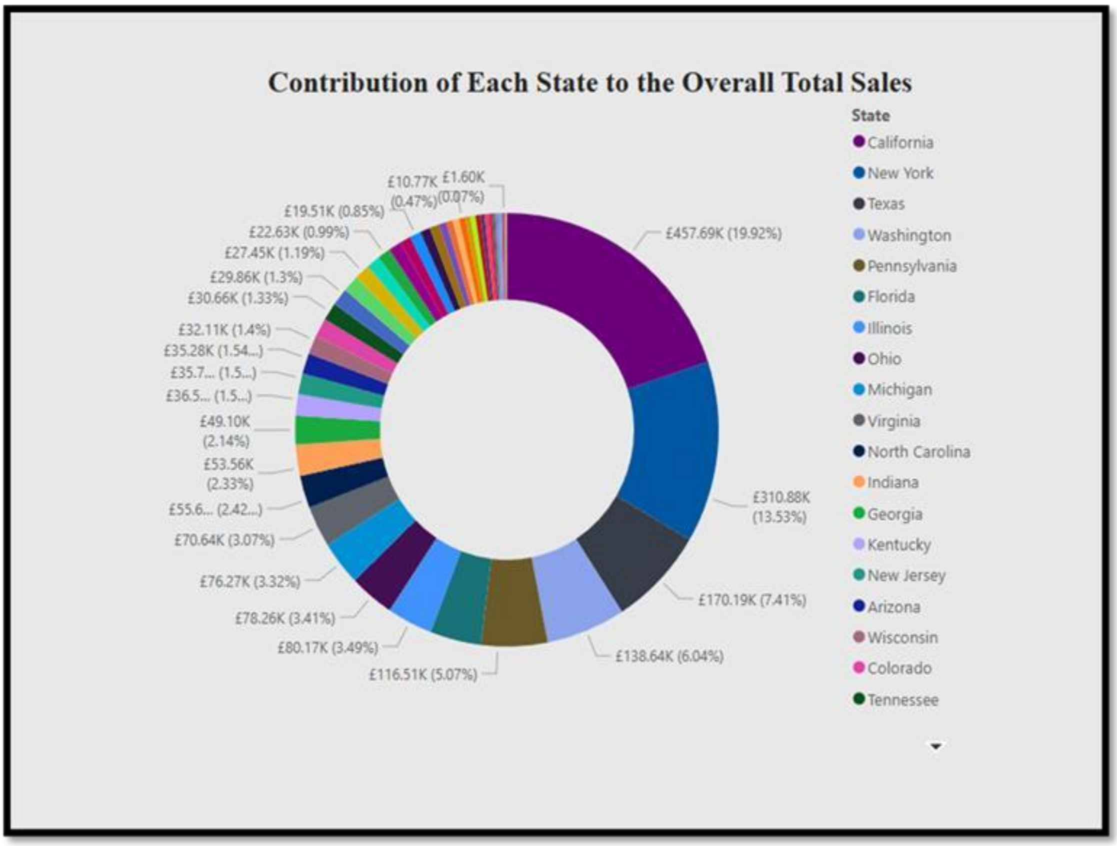
The map visuals depict each of the U.S. state's contributions to Alpha's total sales for 2017. The states are represented by bubbles where size of each bubble reports their sales volume, with larger bubbles representing the highest sales volumes. From the visuals, California, New York, and Washington DC were among the highest contributors based on the fairly large circles. California alone contributed just over £146,000 and represents 6% of total sales. Smaller states represented very small sales and suggested expansion opportunities. Alpha can use this information to allocate resources more efficiently, market them regionally and plan inventory to capitalize on strong markets.



5.6 Total Sales Contributions by State

The donut chart reflects the contributions of each state to Alpha's overall total sales. It highlights that a number of states are far more dominant within the overall sales chart.

California leads by a healthy margin, with 20% of total sales (£457.69K), followed by New York (13.53%), and Texas (7.41%). Other states contribute either below 2%, to somewhere between 2% and 5%. The states like Washington, Pennsylvania or Florida contribute at a higher range beyond 5%. This indicates that Alpha's revenue is generally concentrated in a few states. It may have considerable diversified potential and an immediate growth opportunity to consider updating its strategy in various otherwise under-performing or insignificant areas.



5.7 Comparison of City Sales from 2014 to 2017

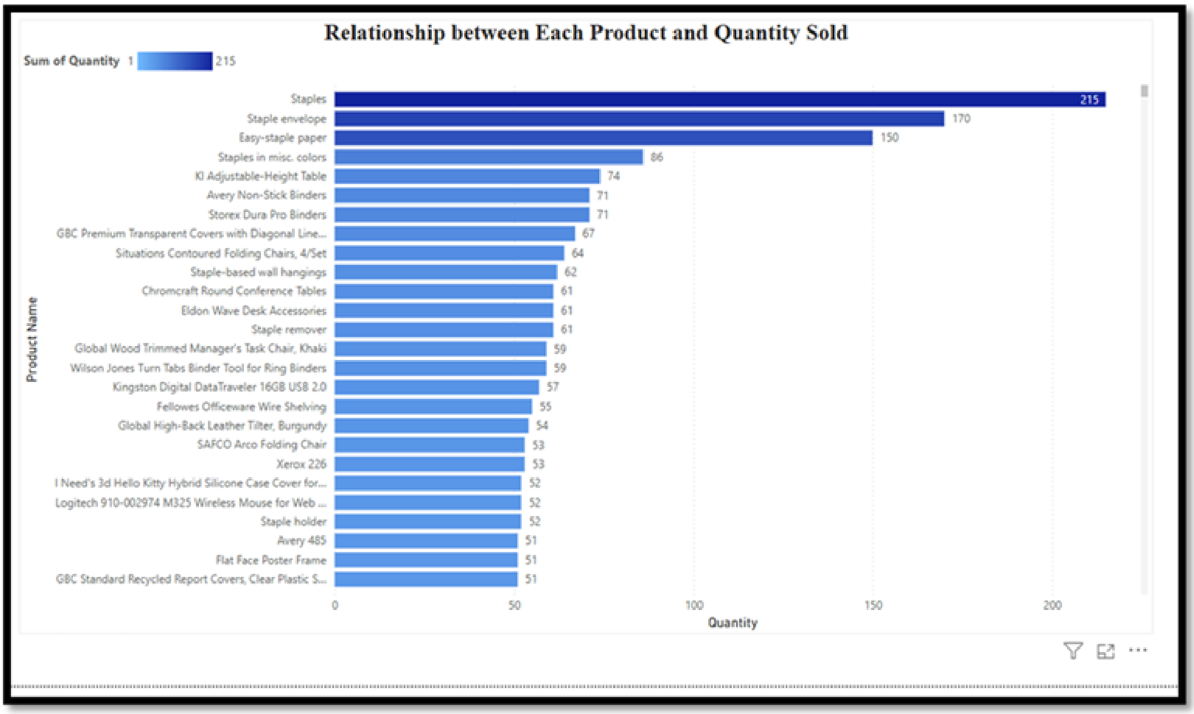
The visual comparison of the total sales per city in 2014 to 2017 highlighted some important trends. First, there appears to be more variation in the sales performance across cities, with Wilmington, Arlington, Atlanta generating high sales over the years. Wilmington has a cumulative total at over £27,000 making it a good market for Alpha. Several cities such as Antioch and Abilene recorded negligible sales having limits to reach customer demand or limited demand. Additionally, the data shows that the majority of the cities on the visual increase in sales from 2014 to 2017 with 2017 having averaged the highest sales. This indicates to a certain degree that a positive growth in business performance from 2014 to 2017 can guide future regional sales discussions and resource allocations.

Total Sales of Each City Between 2014 to 2017

City	2014	2015	2016	2017	Total
Aberdeen				£25.50	**£25.50**
Abilene				£1.39	**£1.39**
Akron	£170.91	£427.55	£804.23	£1,327.30	**£2,729.99**
Albuquerque	£615.49			£1,604.67	**£2,220.16**
Alexandria	£5,221.09	£192.22	£81.20	£25.06	**£5,519.57**
Allen		£290.21			**£290.21**
Allentown		£837.44	£15.81		**£853.25**
Altoona		£20.45			**£20.45**
Amarillo		£3,024.48	£543.25	£205.33	**£3,773.06**
Anaheim	£1,049.97	£1,336.35	£3,533.79	£2,066.76	**£7,986.87**
Andover			£435.85		**£435.85**
Ann Arbor		£889.27			**£889.27**
Antioch			£19.44		**£19.44**
Apopka	£193.15		£711.40		**£904.55**
Apple Valley	£129.30		£563.92	£1,359.80	**£2,053.02**
Appleton	£21.56		£1,649.75		**£1,671.31**
Arlington	£1,510.69	£6,070.82	£10,903.48	£1,729.55	**£20,214.53**
Arlington Heights			£14.11		**£14.11**
Arvada			£503.40		**£503.40**
Asheville	£24.14		£1,363.96	£87.27	**£1,475.38**
Athens	£1,233.45		£201.27	£286.09	**£1,720.81**
Atlanta	£266.44	£7,755.06	£3,443.69	£5,732.65	**£17,197.84**
Atlantic City			£23.36		**£23.36**
Auburn		£692.19	£1,874.24	£588.74	**£3,155.17**
Aurora	£561.68	£2,149.21	£3,719.98	£5,225.61	**£11,656.48**
Austin	£758.35	£795.29	£1,361.59	£3,142.74	**£6,057.98**
Avondale	£892.57	£54.24			**£946.81**
Bakersfield		£214.62	£87.27	£1,075.40	**£1,377.29**
Baltimore	£651.07	£1,161.99	£1,091.26	£3,252.52	**£6,156.84**
Total	**£484,247.50**	**£470,532.51**	**£609,205.60**	**£733,215.26**	**£2,297,200.86**

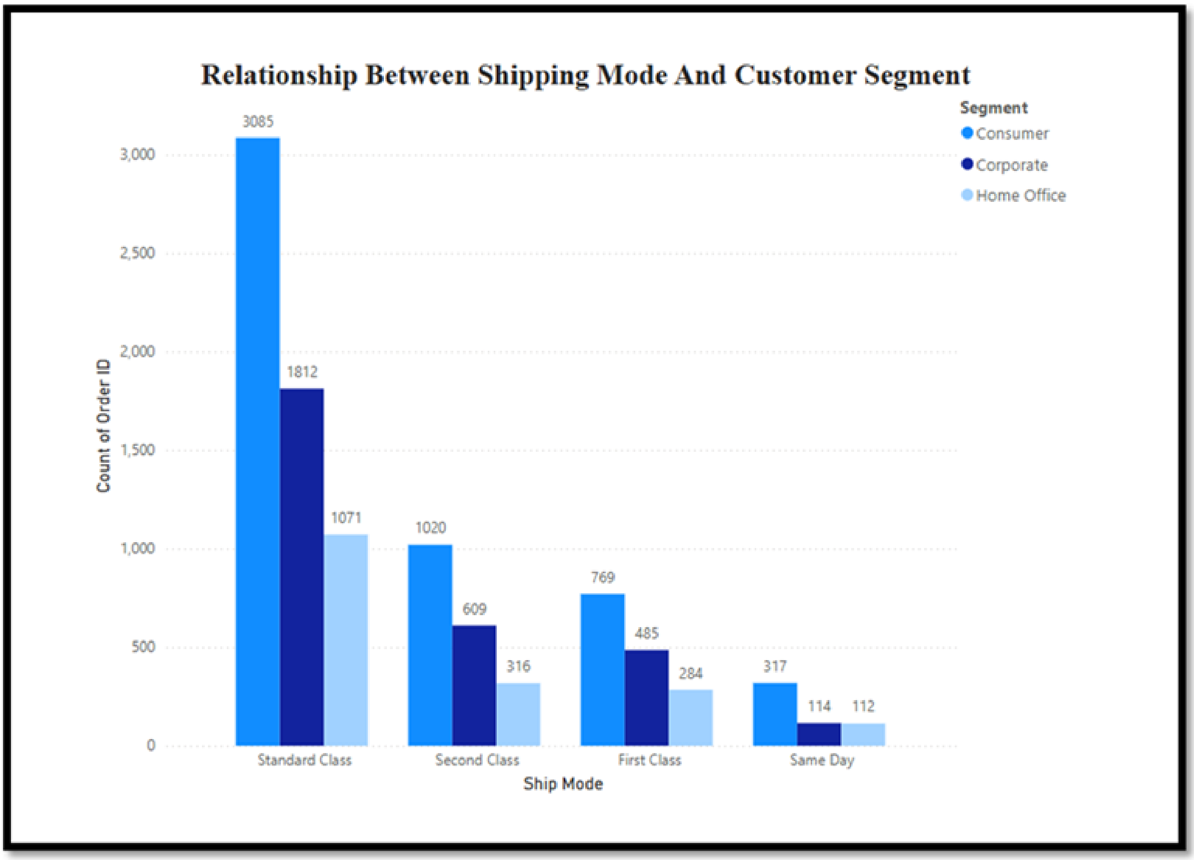
5.8 Relationship Between Product and Quantity Sold

The bar plot showing the relationship between products and quantities sold provides us with further details in the differences across the products. Items such as staples, staple envelopes, and easy-staple paper are evident in their significant quantity sold indicating that they are musthave for everyday business operations. There is also a long tail of products that have very low sales quantities with products sold once or twice. Even though the visualisation shows that a handful of crucial core products account for general high quantity sold, there is also a large amount of very low quantity products. They are most likely contributing to inventory clutter. Considerations could be made based on these value products using the insights gained from the general composition of quantity sold. Some products could be prioritised over others or promoted in conjunction with a high-selling product.



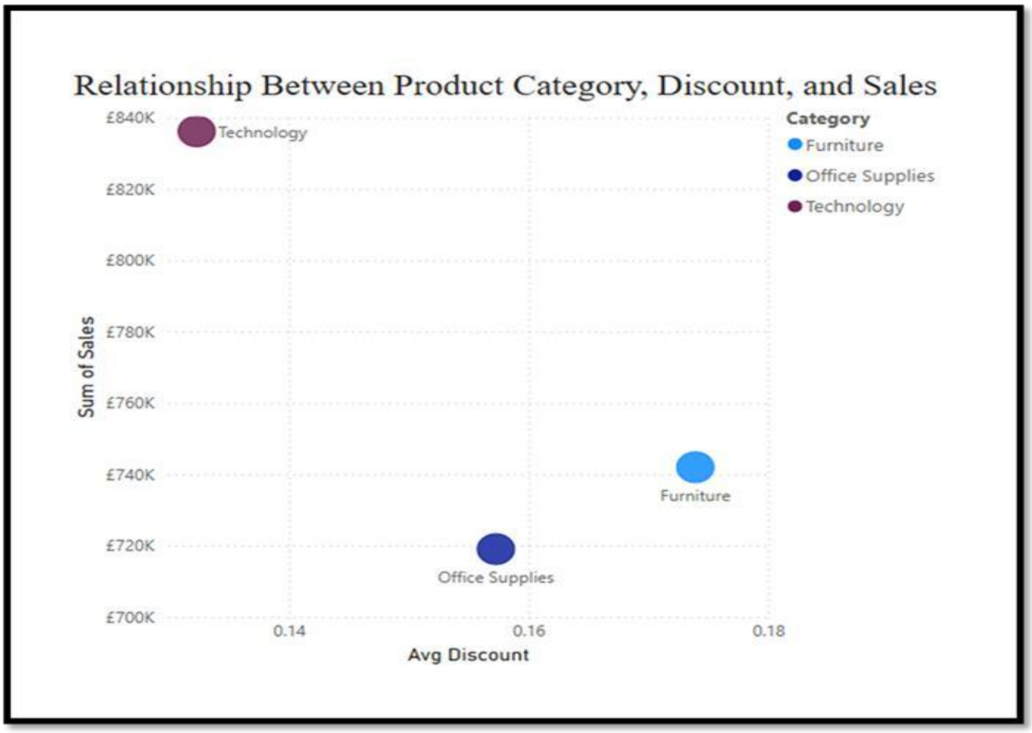
5.9 Relation of Shipping Mode and Customer Segment

A stacked column chart derived from the Order ID count, shows the most common shipping methods per customer segment. The chart shows order shipping modes have a clear preference of the customer segments for Standard Class shipping method where a broad base of consumers have placed over 3,000 orders. This suggests that the 'cost to consumer' may have been a key factor in their decisions. Corporate customers preferred using Standard Class shipping as their most common shipping mode over the other options. Home Office customers and corporate customers ship more using Second Class and First-Class shipping options but this is likely due to 'time required for business' factors. The Same Day shipping mode had the least shipments by volume. It also showed equitable shipments across the segments which suggests the Same Day shipment mode may be used selectively as order shipment with interest relative to urgency not customer segment type. These suggest that Standard Class should continue to be suggested as the default option since it remains the dominant or viable option in face of preference ratings across segments. There is opportunity available to promote advance shipping from corporate clients as they are the most likely group to pay for this delivery preference over other segments types.



5.10 Relationship between Product Category, Sales, and Discount

The bubble chart shows the relationship of average discount against total sales with three types of product categories: Technology, Furniture and Office Supplies. Technology has the highest total sales, but lowest average discount. It is indicative of good demand and customer - perceived value did not require such a large discount to sell the product. Furniture had the largest average discount, and minimal total sales when compared to Technology, demonstrating that just having a high discount does not equate to sales. Office supplies fell somewhere in between in total sales and discount, suggesting a fairly solid, yet not exciting category. So, discounting is a sales signal but even bigger with product type and customer - perceived value. Therefore, a company may want to rethink discounts on low performing categories and increase perceived value marketing on higher performing categories such as Technology which also have lower discounting.



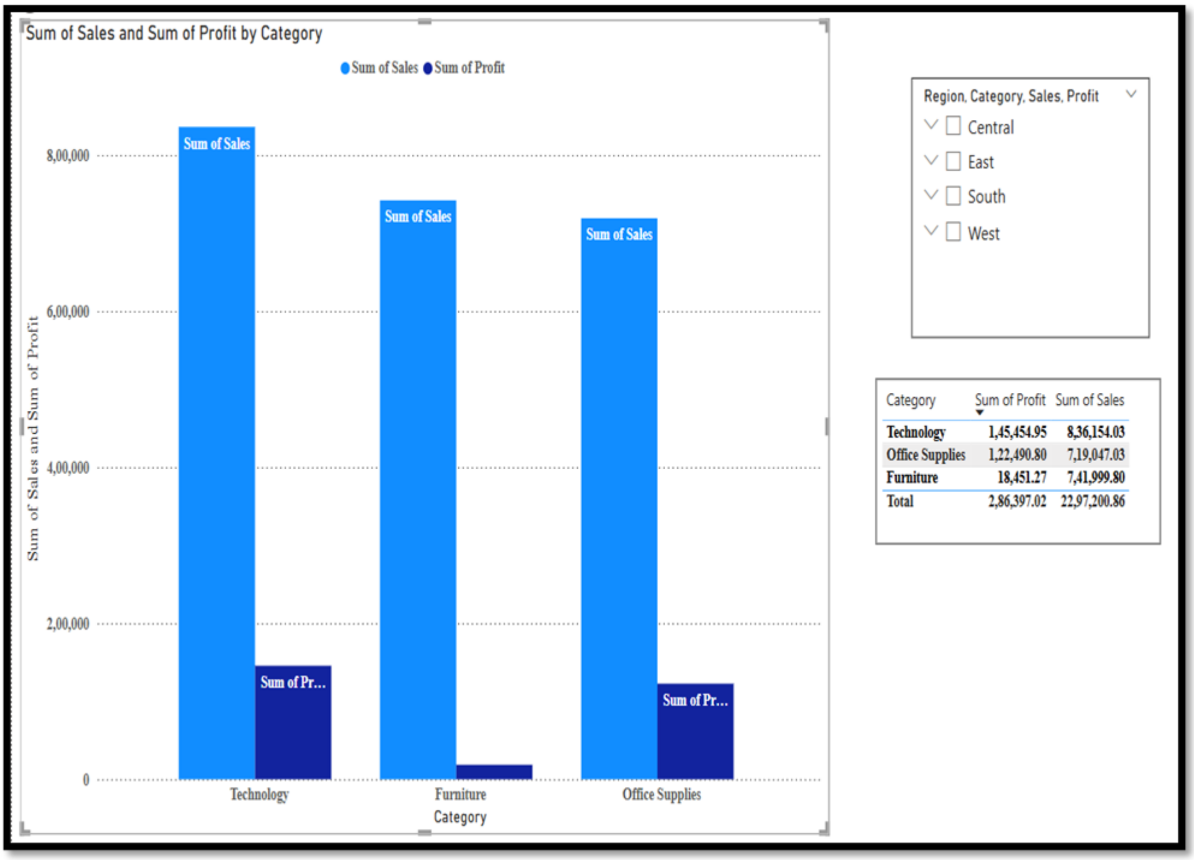
6. Part II Question III

6.1 Alpha Performance Insights Report

This report presents an analysis of Alpha's historical performance in four key business areas: Product Category, Region, State, and Customer Segment. We developed Power BI dashboards using Alpha's transaction data to identify important areas and extract insights that can inform strategic decisions and enhance future sales results.

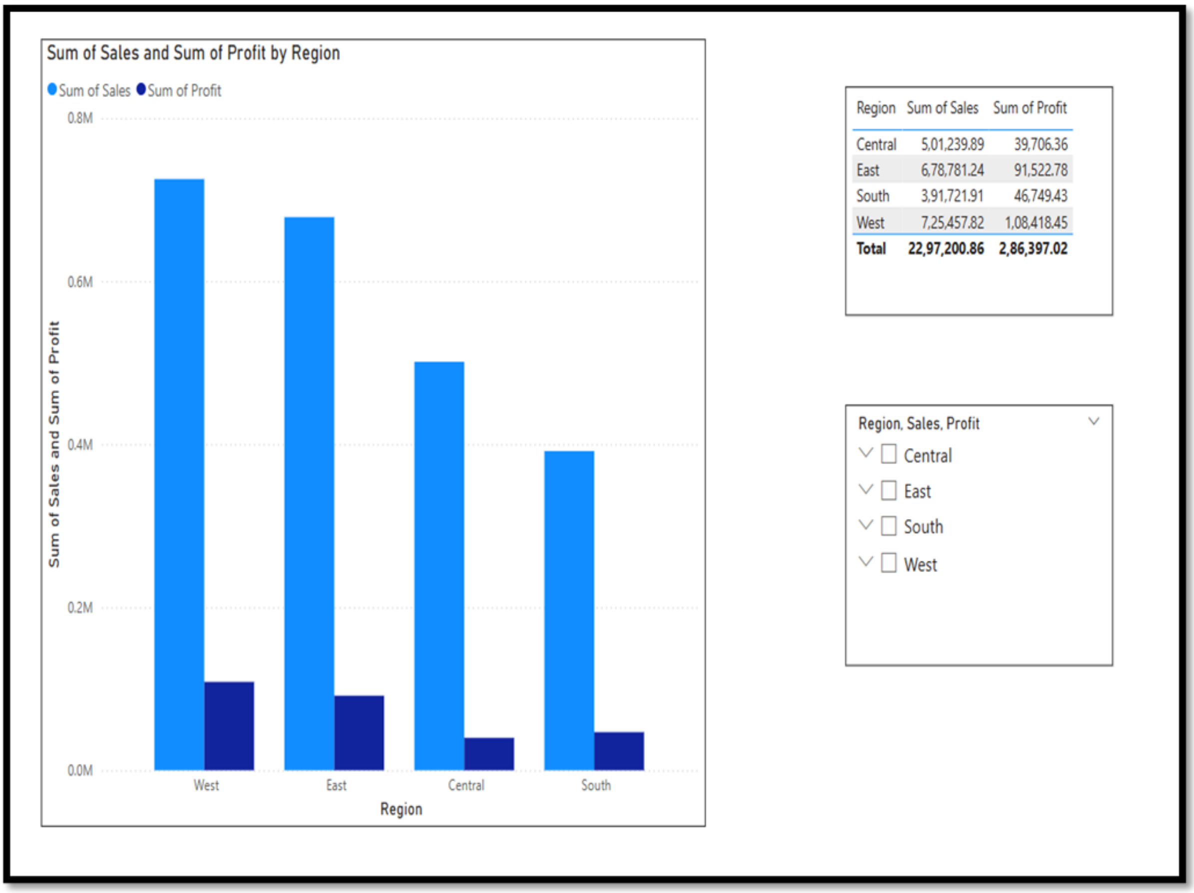
1. Product Category

Our evaluation of Alpha's three primary product categories shows that Technology not only leads in revenue with sales of ₹836 K but also provides the highest profit at ₹145 K. In comparison, Office Supplies generated approximately ₹719 K in sales and contributed a profit of ₹122 K, placing it firmly in the middle range of the pack. Furniture generated ₹742 K in sales but yields next to nothing in profit margin at ₹18 K which may be quite unprofitable. This would indicate that Technology and Office Supplies remain viable opportunities for continued investment through upsell campaigns, premium bundles, or optimizing price because they have differential profit drivers. Whereas, Furniture would require either a price adjustment or a renegotiation of supplier costs to improve upon a very small margin.



2. Region

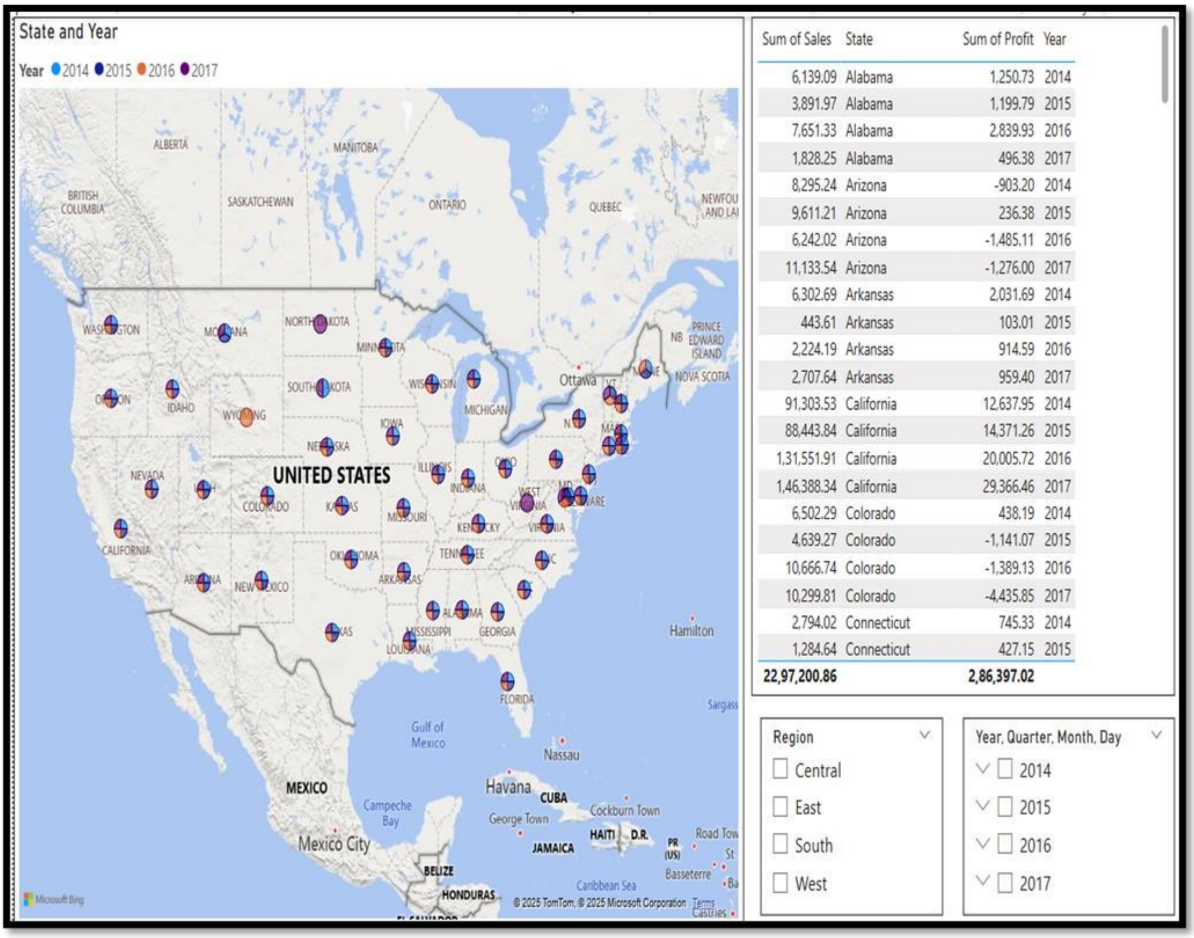
Analysing the sales figures for Alpha's West and East regions reveals that they are the frontrunners, with sales reaching approximately ₹725 K and ₹679 K, while their profits stand at around ₹108 K and ₹91 K, respectively. Meanwhile, the Central region sits in the somewhat modest range of ₹501 K in sales, ₹40 K in profit. South is really inconsistent and underperforming with ₹392 K in sales and ₹47 K in profits. Such disparity underscores the need of codify and test the best practices of the West and East regions in Central and South, such as marketing mix, channel partnerships, or local promotions. Alongside this, a focused turnaround plan needs to be drawn up for South, based on competitive analysis and local offers, to bridge out the performance gap.



Region	Sum of Sales	Sum of Profit
Central	5,01,239.89	39,706.36
East	6,78,781.24	91,522.78
South	3,91,721.91	46,749.43
West	7,25,457.82	1,08,418.45
Total	**22,97,200.86**	**2,86,397.02**

3. State

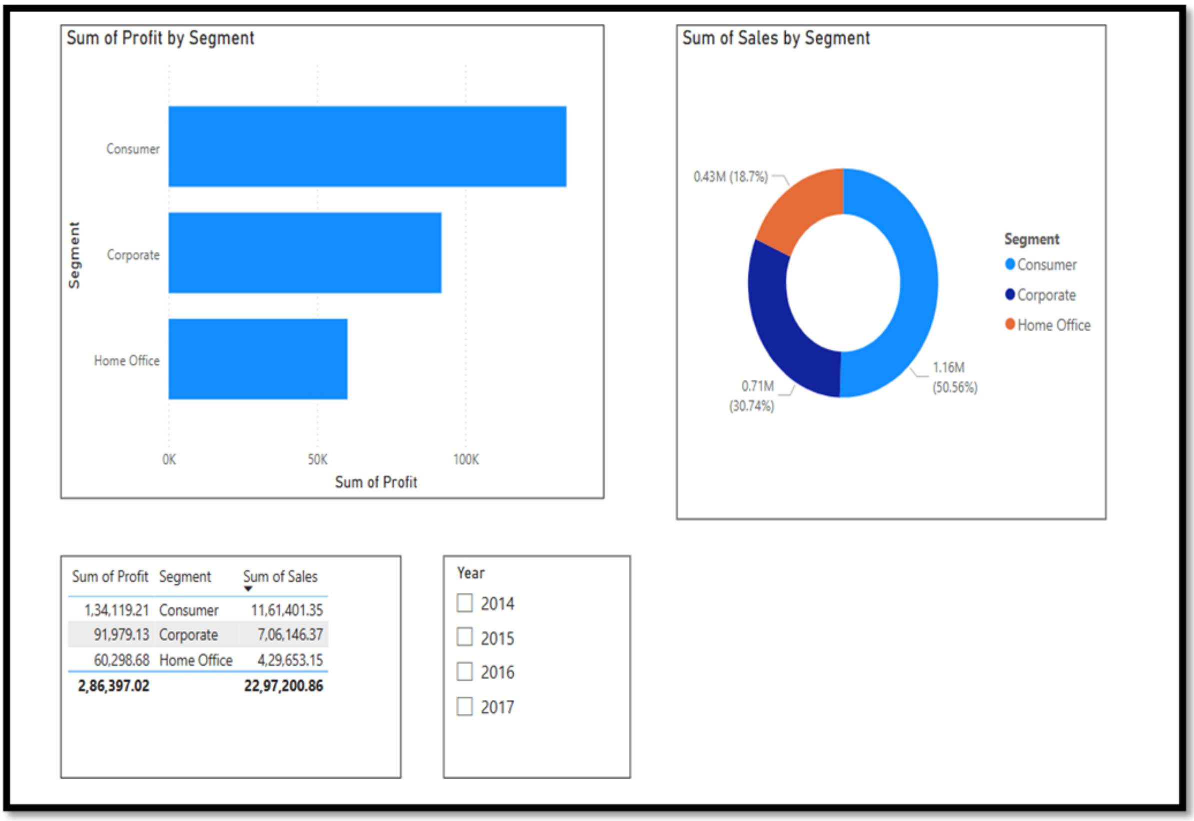
Drilling into the state level data from 2014 to 2017 reveals that powerhouse markets such as California and New York exhibit sales growth and profits steadily year after year affirming their strategic importance. Smaller or emerging markets are volatile which shows that in some years, profit recording a slight negativity suggesting vulnerability to fluctuations or unsuitable product assortments. Alpha should prioritize additional marketing and inventories for states that are consistently growing and conduct a root-cause analysis in the volatile markets. It will test whether markets with varied growth become rapid product markets or whether a strategic withdrawal is more appropriate if foreseeable sustained growth seems unlikely.



Sum of Sales	State	Sum of Profit	Year
6,139.09	Alabama	1,250.73	2014
3,891.97	Alabama	1,199.79	2015
7,651.33	Alabama	2,839.93	2016
1,828.25	Alabama	496.38	2017
8,295.24	Arizona	-903.20	2014
9,611.21	Arizona	236.38	2015
6,242.02	Arizona	-1,485.11	2016
11,133.54	Arizona	-1,276.00	2017
6,302.69	Arkansas	2,031.69	2014
443.61	Arkansas	103.01	2015
2,224.19	Arkansas	914.59	2016
2,707.64	Arkansas	959.40	2017
91,303.53	California	12,637.95	2014
88,443.84	California	14,371.26	2015
1,31,551.91	California	20,005.72	2016
1,46,388.34	California	29,366.46	2017
6,502.29	Colorado	438.19	2014
4,639.27	Colorado	-1,141.07	2015
10,666.74	Colorado	-1,389.13	2016
10,299.81	Colorado	-4,435.85	2017
2,794.02	Connecticut	745.33	2014
1,284.64	Connecticut	427.15	2015
22,97,200.86		**2,86,397.02**	

4. Customer Segment

Segment analysis shows that the Consumer cluster contributes just over fifty percent to total sales (₹1.16 M), and almost a similar percentage to total profit (₹134 K).Hence being the major driving force behind the Alpha's bottom line. The Corporate segment (₹706 K in sales and ₹92 K in profit) and the Home Office segment (₹430 K in sales and ₹60 K in profit) accounts for the remainder of the shares. These numbers give strong credence to the need to bolster Consumer loyalty through retention initiatives programs, while also sourcing in value-added bundles for Corporate and Home Office clients. Interspersing volumes discounts with analytics or service agreements is needed to boost the margin contributions higher up.


Sum of Profit by Segment
Consumer
Corporate
Home Office
Segment
0K
50K
100K
Sum of Profit
Sum of Sales by Segment
0.43M (18.7%)
0.71M (30.74%)
1.16M (50.56%)
Segment
Consumer
Corporate
Home Office
Sum of Profit Segment Sum of Sales
1,34,119.21 Consumer 11,61,401.35
91,979.13 Corporate 7,06,146.37
60,298.68 Home Office 4,29,653.15
2,86,397.02 22,97,200.86
Year
2014
2015
2016
2017

6.2 Return Trends Analysis and Recommendations

This review focuses on product return behaviour from several key dimensions, including category, sub-category, trends over time, customer segments, geographic location, and shipping types.

1. Overall Returns Performance

The benchmark for return rate is set at 5% which reflects industry averages benchmark for consumer goods and retail industries. The key KPI card in our dashboard reflects the Total Return Rate of 8.0%, which exceeds our target of 5% benchmark which was set. When return rates are too high, it indicates a potential problem with product quality management, customer expectations management or fulfilment management.

2. Return Rates by Category

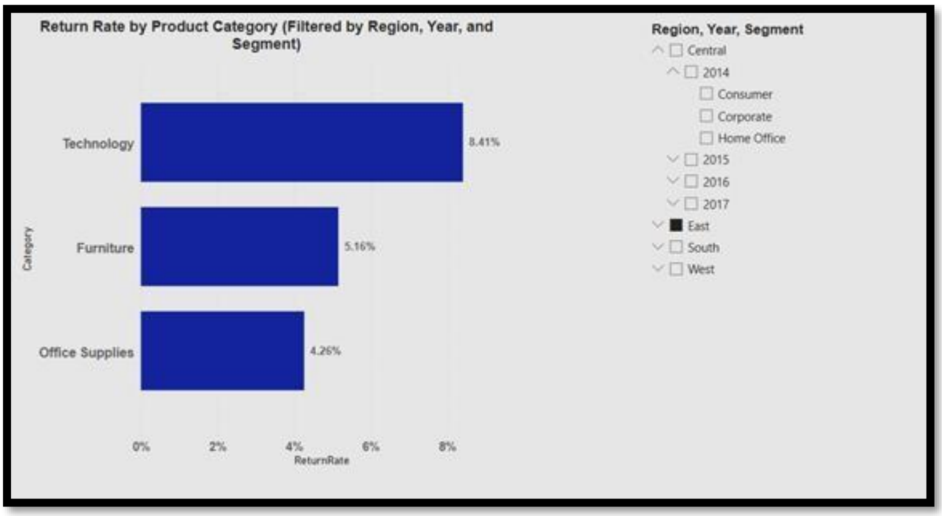
Reviewing performance from a product category standpoint indicates that:

Technology has the highest return rate of 8.45%,

Furniture has a return rate of 8.06%.

Office Supplies has a return of 7.85%.

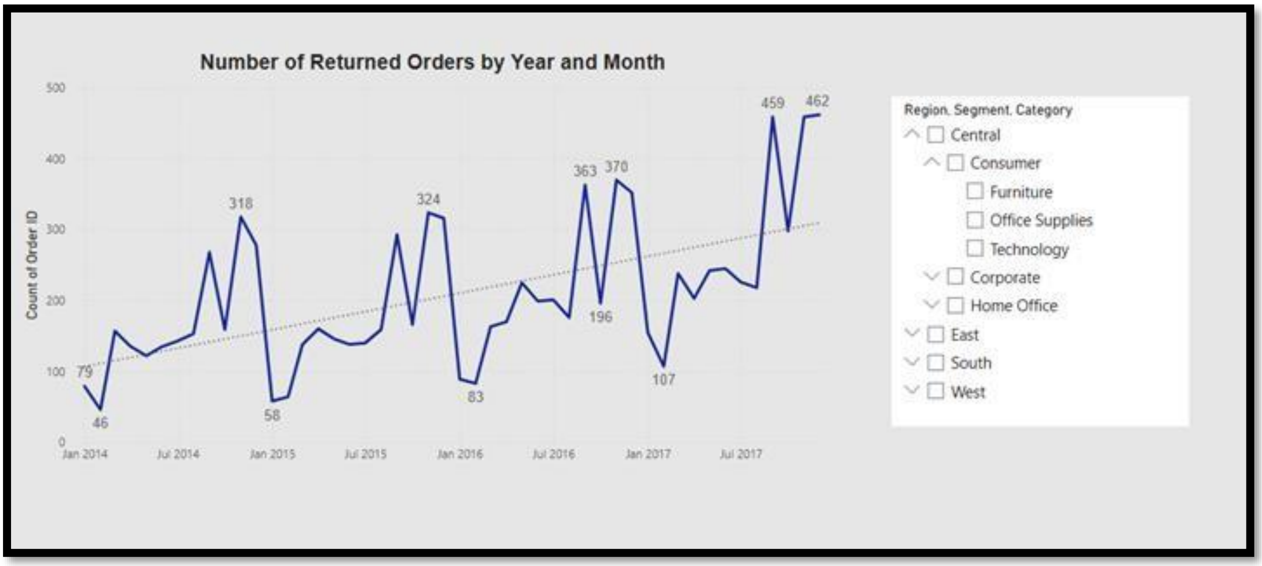
The higher return percentages in the Technology category signals that there are either defects or configuration necessary, or a performance issue that warrants attention. This makes them a priority for improvement.



3. Time Trend of Returns

The line graph represents the Number of Returned Orders over Time (Year-Month), and we can see that the return volumes steadily increased from 2014 - 2017 in an upward trend. There is a monthly peak spike that likely indicates a possible seasonality pattern or promotional influence.

What does this tell us: The upward trend over time may indicate a downward trend for product quality/customer satisfaction or possibly an upward trend of order volume that has never had the proper customer support and/or improvements in Quality Control enacted.



4. **Return Rate by Segment, Sub-Category, and Ship Mode ● Segment:**

Corporate and Consumer segments have higher return rates (8.51% and 8.21%) compared to Home Office (6.56%) return.

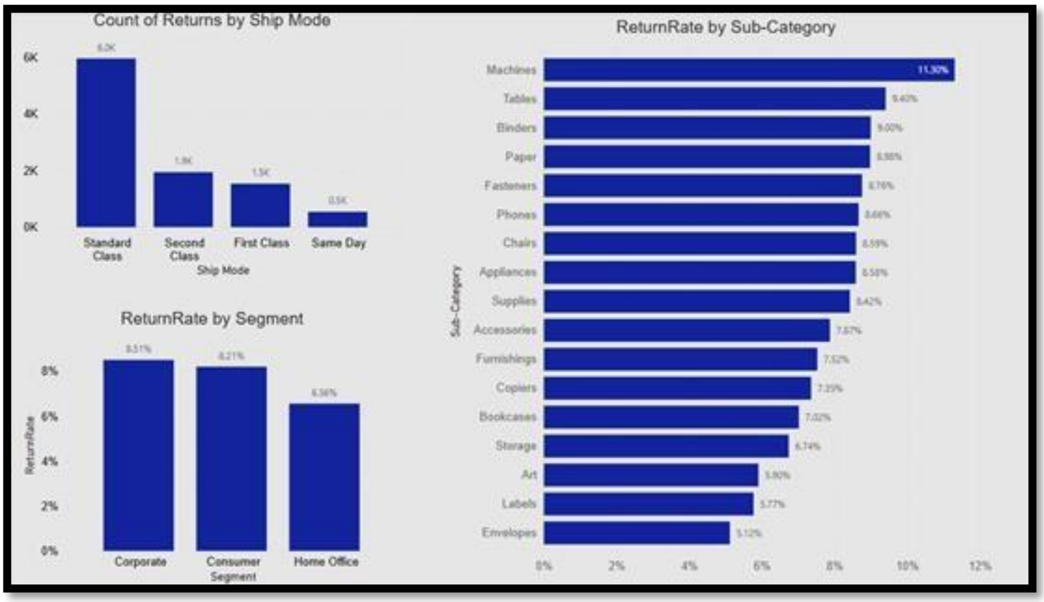
- **Sub-Category:**

Machines (11.30%), Tables (9.40%), and Binders (9.00%) are the sub-categories which are most returned.

- **Ship Mode:**

Standard Class had the greatest number of returns showing strong signs of how speed and handling of shipping may tie into return culture. These visuals allow you to see in detail where the returns are planned, allowing you to target specific products and customers.

These visuals provide a granular view of where returns are most concentrated, helping target specific products and customer groups.

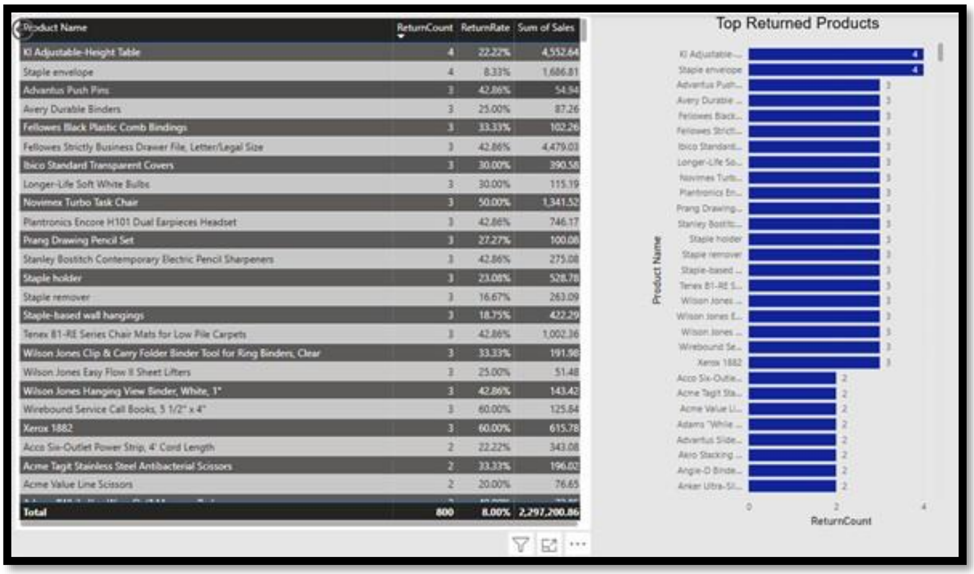


5. Top Returned Products

The table and bar chart highlights:

- Products such as “**KI Adjustable-Height Table**” and “**Staple Envelope**” experience both high returns count and return rates.
- Some high-sellers’ items have low return rates, but others, irrespective of sales, command disproportionately high return percentages.

Hence, the company can set out to examine the outlier products that generally fall out of the pack into peculiar returns needing service quality or customer education.



Product Name	ReturnCount	ReturnRate	Sum of Sales
Kl Adjustable-Height Table	4	22.22%	4,552.64
Staple envelope	4	8.33%	1,686.81
Advantus Push Pins	3	42.86%	54.94
Avery Durable Binders	3	25.00%	87.26
Fellowes Black Plastic Comb Bindings	3	33.33%	102.26
Fellowes Strictly Business Drawer File, Letter/Legal Size	3	42.86%	4,479.03
Ibico Standard Transparent Covers	3	30.00%	390.58
Longer-Life Soft White Bulbs	3	30.00%	115.19
Novimex Turbo Task Chair	3	50.00%	1,341.52
Plantronics Encore H101 Dual Earpieces Headset	3	42.86%	746.17
Prang Drawing Pencil Set	3	27.27%	100.08
Stanley Bostitch Contemporary Electric Pencil Sharpeners	3	42.86%	275.08
Staple holder	3	23.08%	528.78
Staple remover	3	16.67%	263.09
Staple-based wall hangings	3	18.75%	422.29
Tenex B1-RE Series Chair Mats for Low Pile Carpets	3	42.86%	1,002.36
Wilson Jones Clip & Carry Folder Binder Tool for Ring Binders, Clear	3	33.33%	191.98
Wilson Jones Easy Flow II Sheet Lifters	3	25.00%	51.48
Wilson Jones Hanging View Binder, White, 1"	3	42.86%	143.42
Wirebound Service Call Books, 5 1/2" x 4"	3	60.00%	125.84
Xerox 1882	3	60.00%	615.78
Acco Six-Outlet Power Strip, 4' Cord Length	2	22.22%	343.08
Acme Tagit Stainless Steel Antibacterial Scissors	2	33.33%	196.02
Acme Value Line Scissors	2	20.00%	76.65
Total	800	8.00%	2,297,200.86

Based on the insights provided above, it is recommended to undertake the following:

1. **Investigate Categories with High Return Rates**: Concentrate on root cause analyses for the Technology and Machines sub-categories (e.g., quality assurance, incompatibility issues).
2. **Improve Guidance to Customers**: Enhance description with specifications, video demonstrations, and better photographs, especially for technical items.
3. **Shipping Processes**: With standard class shipments, examine enhancements to handling or resetting expectations about delivery times.
4. **Monitoring and Target Setting**: Monitor returns with the Return Rate card and benchmark text ("Target < 5%") to propagate feedback and expectations to internal production teams.
5. **Target Specific Segments Strategies**: Engage Corporate customers to understand bulk return causes and evaluate support or loyalty initiatives to mitigate return frequency.

Conclusion

Return analyses basically provided evidence in some products, segments and categories which are simultaneously causing a contribution in high return rate for the company. By using Power BI , it gives visual support to pinpointing trends between these returns to improve performance, reduce costs, and increase customer satisfaction.

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